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Patents and intellectual property rights

Lecture Scientific Practice

Introduction

Property of movable goods as well as real estate is bound to an object and well regulated by law.

Intellectual property is initially not bound to an object - depending on the type of intellectual property, there are special protective rights:

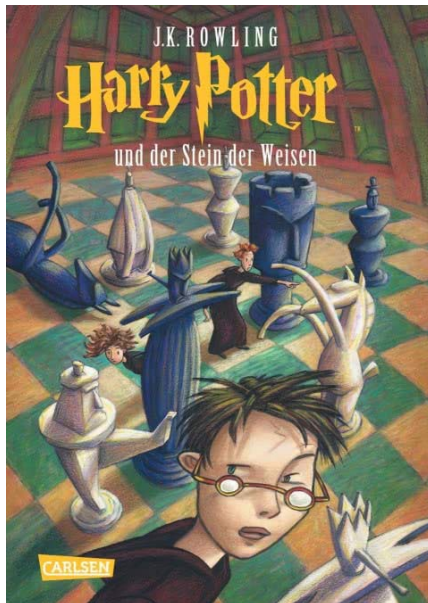
- Copyright
- Trade Marks
- Patents
- Utility Models
- Designs

Copyright

Protection for	works of literature, science and art, as well as photographs and software according to the laws of the respective country
not protectable	official works (e.g. legal texts) works not performed
Term of protection	at least 50 years (in Germany 70 years) after the death of the author
Requirement	originality
Emergence of protection	protection arises automatically when the work is completed (marking with the © symbol)
Utilisation	Copyright collecting societies: GEMA (Germany) for music pieces VG Wort (Germany) for books and magazine articles VG Bild-Kunst (Germany) for reproductions of artworks

Copyright

The Harry Potter example



Source: Carlsen Verlag

Charge brought by Nancy Stouffer as author of "Larry Potter" → dismissed

Countersuit by J.K. Rowlings/Warner Bros. → successful (\$50,000)

Lawsuit by J.K. Rowlings/Warner Bros. against the distribution of the adventures of "Tanja Grotter" & "Harry Potter in Calcutta" & ... → successful

Action by Warner Bros. against fan sites
Boycott by fans against Warner Bros. (PotterWar) → withdrawal of Warner Bros.

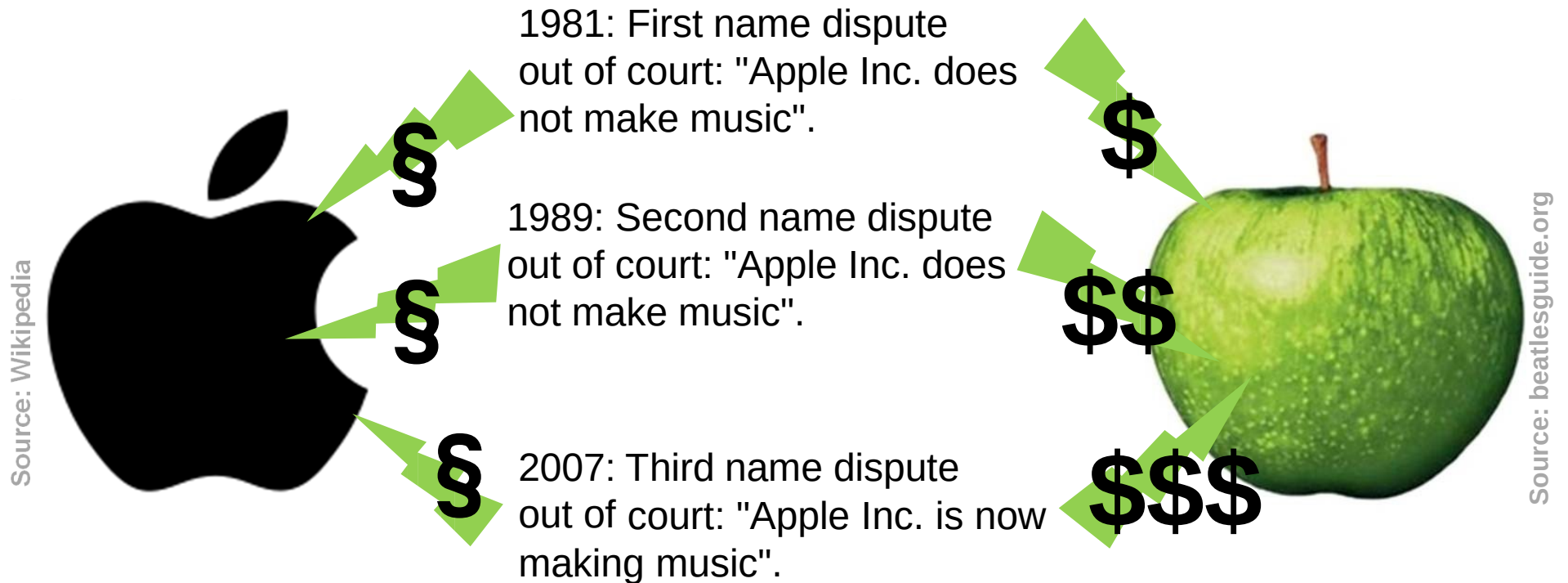
But: no action by J.K. Rowlings/Warner Bros. against parodies like "Barry Trotter and the Shameless Parody" & ff

Trademark law

Protection for	words and figurative signs used for goods, services and works (title protection). Also: geographical indications of source („Champagner“)
not protectable	quality specifications Parle-G Biscuits → Yes Wheat-Biscuits → No
Term of protection	10 years, can be extended as required
Requirement	Distinctive character Use (within 5 years after registration)
Emergence of protection	Registration at the DPMA Marking with the ®-sign or ™ sign
Utilisation	itself

Trademark law

The Apple example

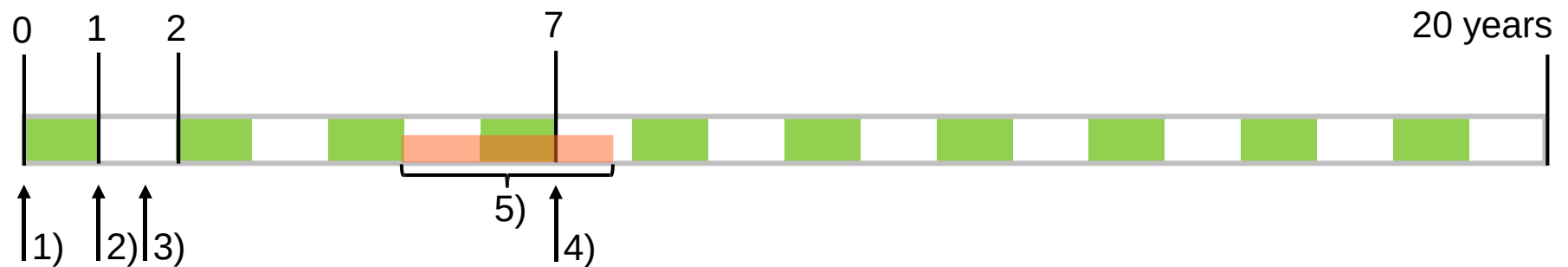


Patent law

Protection for	"Technical ideas" e.g. processes, devices, equipment, constructions, installations, technical products
not protectable	discoveries, scientific theories, medical treatments/surgery, mathematical methods, software
Term of protection	20 years
Requirement	novelty, i.e. does not result from the state of the art at the time of filing ("inventive step") must be based on an inventive step must be commercially viable X perpetuum mobile
Emergence of protection	Granting of a patent by the DPMA
Utilisation	itself

Patent Law

The way to the patent



- 1) Date of filing: decisive for the protective effect, especially for applications with identical contents
- 2) 12-month period: priority period for further developments and international applications
- 3) 18-month period: disclosure by the Patent Office
- 4) 7-year period: Deadline for submission of a examination request
- 5) examination request procedure: If this is successful, the patent can be granted. A 3- month opposition period begins with publication

Patent Law

Costs of a patent procedure

Registration: 40 € (incl. 1st and 2nd annual fee)

Research: 300 €

Examination: 350 €

Annual fee: 70 € (3rd year)

Annual fee: 70 € (4th year)

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Annual fee: 180 € (7th year)

Annual fee: 350 € (10th year)

Annual fee: 1.060 € (15th year)

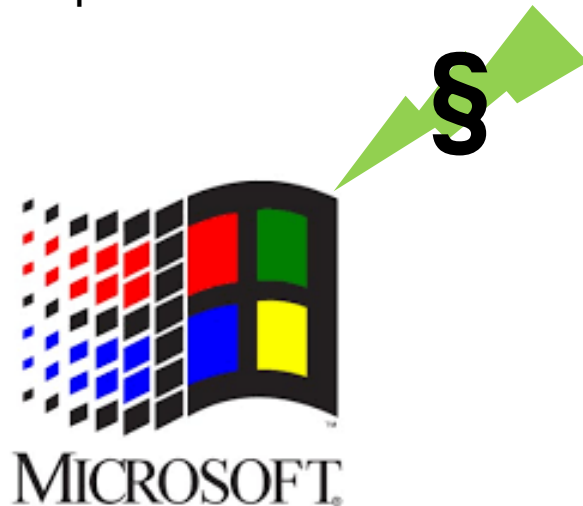
Annual fee: 1.940 € (20th year)

Total: 14.000 € (approx.)

Patent Law

Example GUI (Windows 3.0)

1988: Apple sues Microsoft for the theft of the GUI idea after 6 years rejected by the US Supreme Court



1989: Xerox sues Apple for the theft of the GUI idea after 5 years rejected by the US Supreme Court



Patent Law

Example patent 1)

① **BUNDESREPUBLIK DEUTSCHLAND**

DEUTSCHES PATENTAMT

② **Offenlegungsschrift**
DE 3629434 A1

③ Int. Cl. 4:
H03 M 1/00
H 03 M 7/30

④ Aktenzeichen: P 36 29 434.9
⑤ Anmeldetag: 29. 8. 86
⑥ Offenlegungstag: 3. 3. 88

Veröffentlichung

⑦ **Anmelder:**
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Erlangen, DE

⑧ **Erfinder:**
gleich Anmelder

⑨ **Vertreter:**
Münich, W., Dipl.-Phys. Dr.rer.nat.; Neidl-Stippler,
C., Dipl.-Chem. Dr.phil.nat., Pat.-Anw.; Schiller, W.,
Dr.; Steinmann, O., Dr., Rechtsanw., 8000 München

⑩ **Digitales Codierverfahren**

Beschrieben wird ein digitales Codierverfahren für die Übertragung und/oder Speicherung von akustischen Signalen und insbesondere von Musiksignalen, bei dem N Abtastwerte des akustischen Signals in M Spektralkoeffizienten transformiert werden.
Das erfindungsgemäße Verfahren zeichnet sich dadurch aus, daß die M Spektralkoeffizienten in einer ersten Stufe quantisiert werden. Nach der Codierung wird mit einem Optimalcodierer die zur Darstellung benötigte Bitzahl überprüft. Liegt diese Bitzahl über einer vorgegebenen Bitzahl, so wird die Quantisierung und Codierung in weiteren Stufen so lange wiederholt, bis die zur Darstellung benötigte Bitzahl die vorgegebene Bitzahl nicht überschreitet. Dabei wird zusätzlich zu den Datenbits die benötigte Quantisierungsstufe übertragen bzw. gespeichert.
Das erfindungsgemäße Verfahren hat den Vorteil, daß die Übertragung und/oder Speicherung von akustischen Signalen und insbesondere von Musiksignalen ohne subjektive Qualitätsminderung des Musiksignals bei einer Reduktion der Datenraten um den Faktor 4 bis 6 möglich wird.

DE 3629434 A1

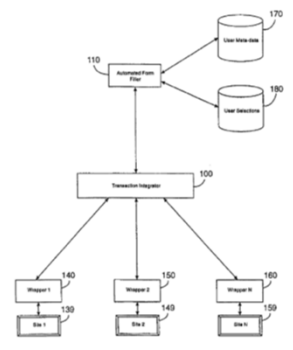
The patent is one of the basic patents for mp3-coding of audio signals.

The patent expired in 2006, so the procedure is in the public domain from that date.

The inventor Karlheinz Brandenburg is now head of the Fraunhofer Institute for Digital Media Technology (Ilmenau).

Patent Law

Example patent 2)

PCT WORLD INTELLECTUAL PROPERTY ORGANIZATION International Bureau	
INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)	
(51) International Patent Classification: G06F 17/60	(11) International Publication Number: WO 99/46707 (43) International Publication Date: 16 September 1999 (16.09.99)
(21) International Application Number: PCT/US99/04442 (22) International Filing Date: 1 March 1999 (01.03.99)	(72) Inventors; and (73) Inventors/Applicants (for US only): PHILLIPS, Yan [US/US]; 7100 132nd Pl S.E. #204, Newcastle, WA 98059 (US); MATHUR, Rakesh [IN/US]; 3473 Eucalyptus Terrace, Fremont, CA 94555 (US); RAJARAMAN, Anand [IN/US]; Suite 310, 1250 Oakmead Parkway, Sunnyvale, CA 94043 (US). (74) Agents: DURDH, Paul, A. et al.; Townsend and Townsend and Crew LLP, 8th floor, Two Embarcadero Center, San Francisco, CA 94111-3834 (US).
(30) Priority Data: 60/077,322 9 March 1998 (09.03.98) US 60/077,655 11 March 1998 (11.03.98) US 60/066,523 20 March 1998 (20.03.98) US 09/066,739 20 March 1998 (20.03.98) US	(81) Designated States: AL, AM, AT, AU, AZ, BA, BB, BG, BR, BY, CA, CH, CN, CU, CZ, DE, DK, EE, ES, FI, GB, GR, GE, GH, GM, HR, HU, ID, IL, IN, IS, JP, KE, KG, KP, KR, KZ, LC, LK, LR, LS, LT, LU, LV, MD, MG, MK, MN, MW, MX, MY, NG, NI, NO, NZ, PG, PH, PT, RO, RU, SD, SE, SG, SI, SK, SL, TH, TM, TR, TT, UA, UG, US, UZ, VN, YU, ZW, ARIPO patent (GH, GM, KE, LS, MW, SD, SI, SZ, UG, ZW), Eurasian patent (AM, AZ, BY, KG, KZ, MD, RU, TJ, TM), European patent (AT, BE, CH, CY, DE, DK, ES, FI, FR, GB, GR, IE, IT, LU, MC, NL, PT, SE), OAPI patent (BF, BI, CF, CG, CI, CM, GA, GN, GW, ML, MR, NE, SN, TD, TG).
(63) Related by Continuation (CON) or Continuation-in-Part (CIP) to Earlier Applications 60/077,322 (CON) US Filed on 9 March 1998 (09.03.98) US 60/077,655 (CON) Filed on 11 March 1998 (11.03.98) US 09/066,523 (CON) Filed on 20 March 1998 (20.03.98) US 09/066,739 (CON) Filed on 20 March 1998 (20.03.98)	(71) Applicant (for all designated States except US): AMAZON.COM, INC. [US/US]; Suite 3600, 900 Fourth Avenue, Seattle, WA 98164 (US).
Published With international search report.	
(54) Title: METHOD AND SYSTEM FOR INTEGRATING TRANSACTION MECHANISMS OVER MULTIPLE INTERNET SITES	
(57) Abstract A method for effecting transactions across multiple vendors in an integrated environment, wherein the user may purchase each of a plurality of items the user finds independent of the vendors. The user's selections (180) are received from the user and supplied to selected set of vendors. When the user is finished, she invokes a check-out application to fill in one or many order entry forms (140) for each of the relevant vendors (139) whose goods the user selected during the course of shopping. The check-out application (110) uses common information (170), such as name, address and credit card number, previously provided by the user in order to fill in the order entry forms (140) for each vendor (139) without requiring the user to fill in these forms. Finally, the check-out application (110) tracks confirmation numbers in a common information store.	
	

The patent is the (notorious) one-click patent of the company Amazon. With this patent the company tried to protect the shopping basket function.

The patent was granted in the USA, but rejected in Europe as a so-called trivial patent.

Patent Law

Conclusion

Patents must fully disclose the invention - in the USA, the patent specification must even contain a "guide to action".

Patents protect intellectual property on the one hand, but on the other hand monopolise technical progress. → They are limited to a maximum term of 20 years.

Patents often serve as a "currency" in disputes between technology companies.

Patent Law

Employee Invention Act (Germany)

The law regulates the rights and obligations of employees and employers in relation to inventions. A distinction is made between service inventions and free inventions. A service invention exists:

- ➡ The invention must have been made during the period of employment.
- ➡ The invention must result from tasks assigned to the employee.

or

The invention must be significantly based on experience or preliminary work of the company.

Patent Law

Service inventions

The employer can claim a service invention.

Obligations of the employee: Immediate, written and fully disclosing invention disclosure to the employer.

Obligations of the employer: Within 4 months after the receipt of the notification, the employer must declare in a written way whether he claims the invention or releases it. In the case of claim, the employee should receive an appropriate remuneration.

Patent Law

Free inventions

If the criteria of a service invention do not apply, the invention is free. Unless the invention is obviously not usable in the employers business:

Obligations of the employee: Immediate, written and partially disclosing invention disclosure to the employer.

The employer must be offered a non-exclusive right of use to reasonable conditions.

Utility model law

Protection for	"Technical ideas"
	e.g. processes , devices, equipment, constructions, installations, technical products
not protectable	discoveries, scientific theories, medical treatments/surgery, mathematical methods, software
Term of protection	10 years
Requirement	novelty, i.e. does not result from the state of the art at the time of filing ("inventive step") compared to the patent, a slightly lower level of invention is allowed
Emergence of protection	grant of a utility model by the DPMA
Utilisation	itself

Design patent law

Protection for not protectable	"outer appearance of objects" e.g. design, surface patterns immovable things (buildings) design elements resulting from the technology (e.g. gear wheel)
Term of protection	20 years, can be extended as required
Requirement	novelty within the cultural area Peculiarity, i.e. the creative individuality must be above the average ability of a designer must be industrially applicable must appeal to the eye must be mobile
Emergence of protection Utilisation	Registration as design at the DPMA itself

Summary

Research

The requested industrial property rights can be searched at the DPMA:

<https://register.dpma.de/DPMAreger/Uebersicht>

European property rights can be searched at the European Patent Office:

<https://worldwide.espacenet.com/>

Summary



Source: www.amazon.de

Patent protection (up to 20 years)

Trademark protection (for any length of time)

Design protection (for any length of time)